

Comprehensive Disease Characterization: Osteogenesis Imperfecta Type VIII

1. Disease Information

Overview

Osteogenesis Imperfecta Type VIII (OI Type VIII) is a severe-to-lethal autosomal recessive form of osteogenesis imperfecta caused by loss-of-function (null) mutations in the *P3H1* gene (formerly known as *LEPRE1*), which encodes prolyl 3-hydroxylase 1. It was first described by Cabral et al. in 2007

([P 17277775](#))

as "a new recessive bone disorder resulting from null LEPRE1 alleles; its phenotype overlaps with lethal/severe osteogenesis imperfecta but has distinctive features." OI Type VIII belongs to the expanding group of recessive OI forms caused by defects in the collagen prolyl 3-hydroxylation complex, which also includes OI Type VII (CRTAP deficiency) and OI Type IX (PPIB/Cyclophilin B deficiency).

Key Identifiers

Database	Identifier
OMIM	#610915 (OI Type VIII); *610339 (P3H1 gene)
MONDO	MONDO:0012581
Orphanet	ORPHA:216796 (within the broader autosomal recessive OI grouping)
ICD-10	Q78.0 (Osteogenesis imperfecta, unspecified — no specific code for Type VIII)
ICD-11	LD24.0 (Osteogenesis imperfecta)
MeSH	D010013 (Osteogenesis Imperfecta)

Synonyms and Alternative Names

- Osteogenesis Imperfecta, Type VIII
- OI Type VIII
- OI8
- P3H1-related osteogenesis imperfecta
- LEPRE1-related osteogenesis imperfecta
- Prolyl 3-hydroxylase 1 deficiency
- Recessive OI due to P3H1 deficiency

Official Disease Description (UniProt)

"A form of osteogenesis imperfecta, a disorder of bone formation characterized by low bone mass, bone fragility and susceptibility to fractures after minimal trauma. Disease severity ranges from very mild forms without fractures to intrauterine fractures and perinatal lethality. Extraskelletal manifestations, which affect a variable number of patients, are dentinogenesis imperfecta, hearing loss, and blue sclerae. OI8 is characterized by disproportionate short stature, shortening of the long bones, white sclerae, a round face and a short barrel-shaped chest." (UniProt Q32P28 disease annotation)

Information Source

Information is derived from aggregated disease-level resources including primary literature (PubMed), OMIM, Orphanet, UniProt, ClinVar, ClinicalTrials.gov, and clinical case series. Individual patient-level data comes from published case reports and cohort studies.

2. Etiology

Disease Causal Factors

Primary cause: Homozygous or compound heterozygous loss-of-function mutations in the *P3H1* gene (chromosome 1p34.2), which encodes prolyl 3-hydroxylase 1. The disease follows strictly autosomal recessive inheritance.

Genetic mechanism: P3H1 is the enzymatic component of the collagen prolyl 3-hydroxylation complex (P3H1/CRTAP/CyPB) in the endoplasmic reticulum. This complex catalyzes 3-hydroxylation of the Pro986 residue on alpha1(I) and alpha1(II) collagen chains. Loss of P3H1 eliminates this modification and disrupts the complex's chaperone function, leading to collagen overmodification, delayed folding, and defective extracellular matrix assembly

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Risk Factors

Genetic Risk Factors

- **Causal variants:** Null mutations in *P3H1* (nonsense, frameshift, splice-site mutations leading to premature termination codons and nonsense-mediated mRNA decay). Most described mutations are loss-of-function

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- **West African founder mutation:** The c.1080+1G>T splice-site mutation is a major population-specific risk allele. Carrier frequency: ~1.48% in Nigeria and Ghana; ~0.4% in Mid-Atlantic African Americans

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This mutation was estimated to have originated 650-900 years before present (1100-1350 CE).

- **Other pathogenic variants reported:** c.628C>T/p.Arg210Ter

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p.Glu351Ter, p.Gly650Arg, c.1720+4G>A

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p.Trp675Leu

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- **Consanguinity:** Significantly increases risk, especially in populations with high carrier frequency. Parental consanguinity documented in 100% of AR OI families in some cohorts

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Environmental Risk Factors

- No specific environmental risk factors have been identified. As a Mendelian genetic disorder, OI Type VIII is fully genetically determined.
- **Trauma:** While not a risk factor for the disease itself, minimal trauma causes fractures in affected individuals due to inherent bone fragility.
- **Age/sex:** No sex predilection (autosomal recessive). Disease manifests at birth or prenatally.

Protective Factors

Genetic Protective Factors

- No specific protective variants have been identified for OI Type VIII.
- Hypomorphic mutations (retaining partial P3H1 function) may lead to milder phenotypes compared to complete null alleles

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- Heterozygous carriers are clinically unaffected.

Environmental Protective Factors

- **Bisphosphonate treatment:** While not preventive, early treatment with bisphosphonates (pamidronate, zoledronic acid) can improve bone mineral density and reduce fracture frequency

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- **Adequate calcium and vitamin D nutrition** may support bone health but does not prevent the disease.

Gene-Environment Interactions

No specific gene-environment interactions have been documented for OI Type VIII. The phenotype is primarily determined by the severity of the *P3H1* mutation (null vs. hypomorphic).

3. Phenotypes

Official HPO Phenotype Annotations (from HPO database, OMIM:610915)

The HPO database contains 29 phenotype annotations for OI Type VIII, all sourced from the seminal paper

([P 17277775](#)).

Below are phenotypes organized by category with documented frequencies:

Skeletal Phenotypes (HPO-curated)

Phenotype	HPO Term	Frequency	Onset	Progression
Multiple prenatal fractures	HP:0005855	5/5 (100%)	Fetal (HP:0011461)	Progressive
Recurrent fractures	HP:0002757	5/5 (100%)	Fetal	May decrease with age/treatment
Osteopenia	HP:0000938	5/5 (100%)	Fetal	Progressive
Thin ribs	HP:0000883	5/5 (100%)	Fetal	Stable; notably without beading
Vertebral compression fracture	HP:0002953	2/2 (100%)	Childhood (HP:0011463)	Progressive
Scoliosis	HP:0002650	1/2 (50%)	Juvenile (HP:0003621)	Progressive
Femoral bowing	HP:0002980	Present	Fetal	Progressive
Tibial bowing	HP:0002982	Present	Fetal	Progressive
Radial bowing	HP:0002986	Present	—	Progressive
Slender long bone	HP:0003100	Present	—	Stable
Barrel-shaped chest	HP:0001552	1/5 (20%)	—	Stable
Platyspondyly	HP:0000926	Present	—	Stable
Kyphosis	HP:0002808	Present	—	Progressive
Femoral retroversion	HP:0008796	1/5 (20%)	—	Stable
Wormian bones	HP:0002645	1/5 (20%)	—	Stable
Type 1 collagen overmodification	HP:0003784	Present	—	—

Growth, Craniofacial, and Limb Phenotypes (HPO-curated)

Phenotype	HPO Term	Frequency	Onset
Disproportionate short-limb short stature	HP:0008873	2/2 (100%)	Congenital
Short metacarpal	HP:0010049	2/2 (100%)	Childhood (HP:0011463)
Wide anterior fontanel	HP:0000260	3/4 (75%)	Neonatal
Decreased calvarial ossification	HP:0005474	2/4 (50%)	Neonatal
Decreased skull ossification	HP:0004331	Present	Neonatal
Round face	HP:0000311	1/4 (25%)	—

Extraskkeletal Phenotypes (HPO-curated)

Phenotype	HPO Term	Frequency	Notes
Proptosis	HP:0000520	1/4 (25%)	Ocular manifestation
Inguinal hernia	HP:0000023	1/5 (20%)	Connective tissue laxity
Joint hypermobility	HP:0001382	1/2 (50%)	Connective tissue laxity
Global developmental delay	HP:0001263	1/2 (50%)	Nervous system
Dentinogenesis imperfecta	HP:0000703	0/2 (0%)	Notably ABSENT — distinguishing from dominant OI
Fetal onset	HP:0011461	5/5 (100%)	Clinical course
Autosomal recessive inheritance	HP:0000007	—	Inheritance

Additional Phenotypes (from broader literature, not in HPO database entry)

Craniofacial Phenotypes

Phenotype	HPO Term	Onset	Frequency
Small to normal head circumference	HP:0000252 (if small)	Neonatal	Variable; distinguishes from dominant OI
Relative macrocephaly (compared to body)	HP:0004482	Childhood	Variable

Extraskkeletal Phenotypes

Phenotype	HPO Term	Onset	Frequency	Notes
White sclerae	HP:0000592	Congenital	>90%	Distinctive from blue sclerae of OI Type I
Hearing loss	HP:0000365	Variable	Probable (confirmed in mouse model; P 23186870)	20-30 dB threshold increase in P3H1 null mice
Bilateral giant retinal tears	HP:0000541	Childhood	Rare (case reports)	Due to P3H1 role in collagen types II and V (P 29329516)
Vascular malformations	HP:0002624	Childhood	Very rare (case reports)	Recently reported (P 42170682)
Dentinogenesis imperfecta	HP:0000703	Childhood	Variable	Less consistently present than in dominant OI

Laboratory Abnormalities

Finding	Type	Notes
Absent/severely reduced $\alpha 1(I)$Pro986 3-hydroxylation (1-4%)	Biochemical	Pathognomonic finding (P 27383115)
Collagen overmodification (excess lysyl hydroxylation)	Biochemical	Increased post-translational modification of collagen helix
Normal bone formation markers	Biochemical	Distinguishes from some other OI forms (P 27383115)
Low areal BMD (Z-score -5 to -6)	Imaging/DXA	Extreme osteopenia
Abnormal collagen fibril morphology	Electron microscopy	Abnormal diameters, irregular borders

Quality of Life Impact

OI Type VIII severely impacts quality of life across all domains: - **Mobility:** Most patients are wheelchair-dependent due to severe skeletal deformity and fracture risk - **Pain:** Chronic bone pain is common; cyclic bisphosphonate therapy reduces pain

([P 30249227](#))

- **Independence:** Severe short stature and skeletal deformity significantly limit self-care and daily activities - **Psychosocial:** Adults with OI experience significant self-perceived burden (SPB) and distress from dependence on caregivers, associated with depression, anxiety, and suicidality

([P 42333980](#))

- **Respiratory:** Respiratory insufficiency is a leading cause of mortality in severe OI. Scoliosis severity negatively correlates with percent-predicted vital capacity. Restrictive pulmonary disease may be due to both vertebral compression fractures/scoliosis and intrinsic pulmonary involvement of abnormal collagen

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- **Quality of life tools:** SF-36 has been used to assess QoL in OI patients; disease-specific tools are lacking

4. Genetic/Molecular Information

Causal Gene

- **Gene symbol:** *P3H1* (previously *LEPRE1*; also known as *GROS1*)
- **HGNC ID:** HGNC:26154
- **NCBI Gene ID:** 64175
- **Chromosomal location:** 1p34.2
- **OMIM gene entry:** *610339
- **Protein:** Prolyl 3-hydroxylase 1 (P3H1); 736 amino acids, 83,394 Da (UniProt Q32P28)
- **EC number:** 1.14.11.7 (procollagen-proline 3-dioxygenase)
- **Signal peptide:** aa 1-22
- **Mature chain:** aa 23-736
- **Key domain:** Fe2OG dioxygenase domain (aa 564-678) — catalytic domain
- **TPR-like helical domain** — mediates protein-protein interactions
- **KDEL ER retention motif:** aa 733-736 — prevents secretion from ER

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- **Cofactors:** Fe²⁺ (CHEBI:29033) and L-ascorbate (CHEBI:29073, vitamin C) — required for catalytic activity
- **Catalytic reaction:** L-prolyl-[collagen] + 2-oxoglutarate (CHEBI:16810) + O₂ → trans-3-hydroxy-L-prolyl-[collagen] (CHEBI:18240) + succinate (CHEBI:15741) + CO₂
- **Complex:** Forms a 1:1:1 PCP complex with CRTAP and PPIB/CyPB; two PCP complexes can further associate to form higher-order assemblies (UniProt, citing

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- **Subcellular location:** Endoplasmic reticulum (primary); also secreted into extracellular matrix

- **InterPro domains:** IPR039575 (P3H), IPR005123 (Oxoglutarate/Fe-dependent dioxygenase), IPR011990 (TPR-like helical domain superfamily), IPR056585 (Leprecan domain)
- **Reactome pathway:** R-HSA-1650814 (Collagen biosynthesis and modifying enzymes)

Pathogenic Variants

ClinVar database: 193 pathogenic variants in P3H1 (out of 874 total variants catalogued), confirming extensive allelic heterogeneity.

Variant types: Predominantly loss-of-function: - **Nonsense mutations:** e.g., c.628C>T (p.Arg210Ter)

(P 34637196);

p.Glu351Ter

(P 41499654);

p.Gln714Ter (ClinVar); p.Tyr304Ter (ClinVar); p.Glu41Ter (ClinVar); p.Gln668Ter (ClinVar); p.Gln689Ter (ClinVar) - **Splice-site mutations:** e.g., c.1080+1G>T (West African founder mutation)

(P 22281939);

c.1720+4G>A

(P 41499654);

c.1170+2T>A; c.1838+1G>T; c.465+2T>G; c.618+1G>A; c.1474-2A>G; c.1720+1G>T; c.2055+1G>C (all in ClinVar) - **Frameshift mutations:** e.g., p.Ala40fs; p.Ala162fs; p.Arg359fs; p.Glu374fs; p.Lys401fs; p.Arg547fs; p.Val688fs; p.Leu693fs (ClinVar) - **Missense mutations (rare but pathogenic):** e.g., p.Gly650Arg

(P 41499654);

p.Trp675Leu (predicted disease-causing;

(P 26634552);

p.Glu573Lys (in catalytic domain; ClinVar); p.Gly449Arg (ClinVar); p.Arg115His (ClinVar); p.Ser291Thr (ClinVar) - **Stop-loss mutation:** p.Ter737ThrextTer? (c.2209_2210del) — extends reading frame past the normal stop codon (ClinVar)

Variant classification: Pathogenic per ACMG/AMP guidelines for established null alleles. Most mutations are classified as pathogenic or likely pathogenic.

Allele frequency: - c.1080+1G>T: Carrier frequency 1.48% in West Africa (Nigeria/Ghana), 0.4% in African Americans

([P 22281939](#))

- Most other variants are ultra-rare/private family mutations with frequencies below detection thresholds in gnomAD

Germline vs. somatic: All known variants are germline.

Functional consequences: Loss of function. Null mutations lead to: 1. Nonsense-mediated mRNA decay with minimal transcript levels 2. Absent or severely reduced P3H1 protein 3. Secondary reduction of CRTAP protein (mutual stabilization;

([P 19846465](#))

4. Loss of Pro986 3-hydroxylation on $\alpha 1(I)$ collagen 5. Collagen overmodification (excess lysyl hydroxylation and glycosylation) 6. Delayed but increased total collagen secretion

([P 17277775](#))

Structural Biology: Cryo-EM Structures of the PCP Complex

Six cryo-EM structures of the human P3H1/CRTAP/PPIB complex were determined in 2024

([P 39245686](#)),

providing the first atomic-resolution view of the disease-relevant collagen processing machinery:

PDB ID	Description	Resolution
8K0E	P3H1/CRTAP heterodimer	3.65 Å
8K0F	P3H1/CRTAP/PPIB heterotrimer (apo state)	3.37 Å
8K0I	P3H1/CRTAP/PPIB heterotrimer (dual-ternary state)	3.62 Å
8K0M	Heterotrimer + 2-oxoglutarate (CHEBI:16810)	3.17 Å
8K17	Heterotrimer + collagen $\alpha 1(I)$ peptide substrate	3.18 Å
8KC9	Heterotrimer + cyclosporin A (CHEBI:4031, CyPB inhibitor)	3.75 Å

Key structural insights: - The active sites of P3H1 (3-hydroxylase) and PPIB (cis-trans isomerase) form a **face-to-face bifunctional reaction center**, indicating a coupled modification mechanism - Multiple collagen binding sites create a substrate interacting zone - An unexpected **dual-ternary complex** (two PCP units) was observed, and the balance between states is altered by active-site mutations - These structures enable atomic-level mapping of OI-causing mutations

Protein Interaction Network (STRING)

P3H1's highest-confidence interaction partners map the collagen modification machinery:

Partner	STRING Score	Role	OI Subtype if Mutated
PPIB (CyPB)	0.999	PCP complex member; peptidyl-prolyl isomerase	OI Type IX
CRTAP	0.999	PCP complex member; helper protein	OI Type VII
COL1A2	0.945	Collagen substrate ($\alpha 2(I)$ chain)	OI Types I-IV
COL1A1	0.945	Collagen substrate ($\alpha 1(I)$ chain)	OI Types I-IV
SERPINH1 (HSP47)	0.911	Collagen chaperone	OI Type X
P4HB	0.875	Prolyl 4-hydroxylase subunit (PDI)	—
FKBP10	0.866	Collagen foldase	OI Type XI / Bruck syndrome
P4HA2	0.827	Prolyl 4-hydroxylase alpha-2	—
TMEM38B	0.796	ER cation channel	OI Type XIV
COL5A1	0.796	Type V collagen (also 3-hydroxylated)	Ehlers-Danlos
COL5A2	0.793	Type V collagen	Ehlers-Danlos
P4HA1	0.774	Prolyl 4-hydroxylase alpha-1	—
P3H3	0.758	Prolyl 3-hydroxylase 3	—
COL2A1	0.745	Type II collagen (also 3-hydroxylated)	Stickler syndrome
PLOD2 (LH2)	0.744	Lysyl hydroxylase 2	Bruck syndrome type 2

gnomAD Gene Constraint Metrics

- **pLI:** 1.22×10^{-19} (NOT loss-of-function intolerant — expected for AR disease gene; heterozygous carriers are unaffected)
- **o/e LoF:** 0.82 (near expected; LoF variants are tolerated in heterozygosity)
- **LoF Z-score:** 1.38
- **Missense Z-score:** 1.17

- **o/e Missense:** 0.91
- **Chromosomal location (GRCh38):** chr1:42,746,335-42,767,084
- **Ensembl ID:** ENSG00000117385

Tissue Expression

- **Human Protein Atlas:** Low tissue specificity — P3H1 is ubiquitously expressed, consistent with collagen's widespread tissue distribution
- P3H1 expression is expected in all collagen-producing tissues including bone, cartilage, skin, tendon, and dentin

LEPRE1 Splice Forms

Three LEPRE1 mRNA splice forms have been identified. The disease-causing splice form encodes the 736 amino acid protein with a KDEL ER retention signal. Splice site mutations may affect only specific isoforms, partially explaining phenotypic variability (P 19088120).

Modifier Genes

No specific modifier genes have been confirmed for OI Type VIII. However: - **CRTAP** and **PPIB** encode the other components of the prolyl 3-hydroxylation complex; variants in these genes could theoretically modify phenotype - Additional variants in collagen genes (e.g., COL5A2) have been identified in compound presentations (P 32770541).

- One study identified digenic or modifier effects with additional AD OI gene variants (P 40650436).

Epigenetic Information

No specific epigenetic modifications have been reported for OI Type VIII. General OI research suggests potential roles for DNA methylation changes in osteoblast differentiation, but this remains unexplored for P3H1-related OI.

Chromosomal Abnormalities

OI Type VIII is caused by point mutations or small insertions/deletions, not large-scale chromosomal abnormalities.

5. Environmental Information

Environmental Factors

OI Type VIII is a purely genetic disorder. No environmental factors contribute to disease causation. However, environmental factors influence disease severity: - **Trauma:** Even minimal trauma causes fractures in affected individuals - **Immobilization:** Prolonged immobility can worsen bone loss

Lifestyle Factors

- **Physical activity:** Carefully supervised physical activity and physiotherapy are important for maintaining bone density and muscle strength
- **Nutrition:** Adequate calcium and vitamin D intake supports bone health

Infectious Agents

Not applicable. OI Type VIII is not caused by or triggered by infectious agents.

6. Mechanism / Pathophysiology

Molecular Pathways

The pathogenic cascade in OI Type VIII involves:

1. **Loss of the prolyl 3-hydroxylation complex (Primary defect)** - P3H1 null mutations → absence of P3H1 protein - Secondary loss of CRTAP via mutual destabilization (proteasomal degradation;

- CyPB levels unaffected but loses collagen interaction
([P 19997487](#))
- **KEGG pathway:** Protein processing in endoplasmic reticulum (hsa04141) - **Reactome:** Collagen biosynthesis and modifying enzymes (R-HSA-1650814)
- 2. Defective collagen post-translational modification** - Loss of 3-hydroxylation of $\alpha 1(I)$ Pro986 (reduced to 1-4%;
([P 27383115](#))
- Excess lysyl hydroxylation and glycosylation of the collagen helix (overmodification;
([P 17277775](#))
- This overmodification indicates delayed triple helix formation
- 3. Impaired collagen folding and ER processing** - Delayed collagen folding in the ER - Partial ER retention of overmodified procollagen, causing enlarged ER cisternae
([P 32173581](#))
- Paradoxically increased total collagen secretion
([P 17277775](#))
- **GO terms:** GO:0030235 (nitric-oxide synthase regulator activity — related to ER stress); GO:0034975 (protein folding in endoplasmic reticulum)
- 4. Abnormal collagen fibril assembly** - Secreted collagen has abnormal structure - Collagen fibrils show abnormal diameters and irregular borders
([P 27383115](#))
- Disorganized extracellular fibers
([P 32173581](#))
- 5. Defective bone matrix and mineralization** - Decreased cortical width and very thin trabeculae - Patches of increased osteoid - Paradoxically increased matrix mineralization
([P 27383115](#))
- Increased proportion of low-mineralization bone - Abnormal cross-linking chemistry
([P 23508630](#))

Cellular Processes

- **ER stress / Unfolded Protein Response (UPR):** Collagen misfolding and ER retention may trigger UPR signaling pathways including PERK, ATF6, and IRE1
([P 33798677](#));
([P 32980496](#))
- **Proteasomal degradation:** CRTAP protein is degraded via proteasome when P3H1 is absent
([P 19846465](#))
- **Osteoblast dysfunction:** Altered intracellular homeostasis and ECM deposition
([P 32980496](#))
- **GO terms:** GO:0030574 (collagen catabolic process); GO:0006457 (protein folding); GO:0030199 (collagen fibril organization)

Protein Dysfunction

- **P3H1 loss of function:** Complete absence of enzymatic activity (prolyl 3-hydroxylase) and chaperone function
- **Collagen structural alteration:** Overmodified collagen helix with abnormal post-translational modifications
- Studies using a P3H1 catalytic-dead knock-in mouse (Lepre1^{H662A}) revealed that loss of 3-hydroxylation alone (with retained chaperone function) produces a milder phenotype, indicating the chaperone function is critical
([P 24465224](#)).

As stated: "The relative contribution of losing this complex's 3-hydroxylation versus PPIase and collagen chaperone activities to the phenotype of recessive OI is unknown."

Critical mechanistic insight — chaperone vs. enzymatic function: A knock-in mouse with an $\alpha 1(I)P986A$ substitution (cannot be 3-hydroxylated, but retains the PCP complex) showed "normal survival, growth, femoral breaking strength and mean bone mineralization. However, the bone collagen HP/LP crosslink ratio is nearly doubled in mutant mice, while collagen fibril diameter and bone yield energy are decreased. Thus, 3-hydroxylation of the A1 site $\alpha 1(I)P986$ affects collagen crosslinking and structural organization, but its absence does not directly cause recessive bone dysplasia"

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This proves that the severe phenotype of OI Type VIII arises primarily from loss of the PCP complex's chaperone/foldase function, NOT merely from loss of the 3-hydroxylation modification.

Role of 3-hydroxyproline in fibril assembly: The 3Hyp residues at Pro986 are "positioned within mutually interactive binding motifs on adjacent collagen molecules that contribute through hydrogen bonding to the process of fibril supramolecular assembly"

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Loss of 3Hyp alters collagen cross-link chemistry (HP/LP ratio nearly doubled) and fibril diameter; providing a subtle but measurable structural defect on top of the dominant chaperone-loss phenotype.

Metabolic Changes

- **Collagen metabolism:** Altered post-translational modification pathway
- **Bone turnover:** Normal bone formation markers despite severe osteopenia

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- **Calcium/phosphate metabolism:** Secondary effects from skeletal fragility

Immune System Involvement

Not a primary feature. No autoimmune or inflammatory components have been described.

Tissue Damage Mechanisms

- **Mechanical fragility:** Abnormal collagen fibrils reduce bone mechanical strength
- **Growth plate disruption:** Altered endochondral ossification leads to growth deficiency
- Bulbous epiphyseal deformity and popcorn calcifications may occur, reflecting growth plate differentiation abnormalities

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Biochemical Abnormalities

- **Enzyme deficiency:** Prolyl 3-hydroxylase 1 activity absent

- **Substrate accumulation:** Non-hydroxylated Pro986 residue on $\alpha 1(I)$ collagen
- **Product deficiency:** 3-hydroxyproline at position 986 reduced to 1-4%
- **Secondary overmodification:** Excess lysyl hydroxylation reflecting delayed folding

Signaling Pathways Affected

According to comprehensive review (

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- **WNT signaling** (GO:0016055) - **RANK/RANKL signaling** (bone resorption regulation) - **TGFbeta signaling** (GO:0007179) — identified as the top activated signaling pathway in OI bone via GSEA; SMAD phosphorylation most significantly upregulated molecular event; TGF-beta1 identified as most significant activated upstream regulator

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- **MAPK signaling** (GO:0000165) - **Integrin-mediated signaling** (GO:0007229) - **Unfolded protein response** (GO:0030968)

Mechanistic Causal Chain: From Gene to Disease

The pathogenesis of OI Type VIII can be summarized as a step-wise causal cascade:

P3H1 null mutation (biallelic)

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P3H1 protein absent from ER

→ CRTAP degraded via proteasome (mutual stabilization lost; <https://pubmed.ncbi.nlm.nih.gov/25411111/>)

↓
PCP complex (P3H1/CRTAP/CyPB) abolished

→ Loss of collagen CHAPERONE/FOLDASE function ← PRIMARY DRIVER

→ Delayed procollagen triple helix folding

↓
Excess lysyl hydroxylation & glycosylation (OVERMODIFICATION)

↓
Partial ER retention → enlarged ER cisternae → UPR activation

→ Paradoxically increased but ABNORMAL collagen secretion

↓
Disorganized collagen fibrils (abnormal diameter/borders)

↓
Defective bone matrix assembly

→ Loss of Pro986 3-HYDROXYLATION ← SECONDARY MODIFIER

→ Altered collagen cross-linking (HP/LP ratio doubled; <https://pubmed.ncbi.nlm.nih.gov/25411111/>)

→ Disrupted intermolecular H-bonding in fibril assembly (<https://pubmed.ncbi.nlm.nih.gov/25411111/>)

↓
Subtle structural defect (insufficient alone to cause OI)

↓
COMBINED EFFECT: Severe bone fragility

→ Decreased cortical width & very thin trabeculae

→ Paradoxically increased matrix mineralization

→ Recurrent fractures from minimal trauma

→ Growth plate disruption → severe short stature

→ Progressive skeletal deformity (scoliosis, bowing)

→ Respiratory compromise from thoracic deformity → potential lethality

Key evidence distinguishing chaperone from enzymatic function: - P3H1 null (no complex)

→ severe/lethal OI

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- P3H1 catalytic-dead knock-in (complex intact, no hydroxylation) → milder phenotype

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- $\alpha 1(I)$ P986A knock-in (complex intact, substrate unmodifiable) → NO bone dysplasia

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- Zebrafish (naturally lack 3Hyp) + p3h1 knockout → OI phenotype from chaperone loss alone

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[P 32173581](#)

P3H1-Specific GO Annotations (from UniProt Q32P28)

Molecular Function: - GO:0019797 — procollagen-proline 3-dioxygenase activity - GO:0005506 — iron ion binding - GO:0031418 — L-ascorbic acid binding

Biological Process: - GO:0060348 — bone development - GO:0032963 — collagen metabolic process - GO:0006457 — protein folding - GO:0018126 — protein hydroxylation - GO:0050821 — protein stabilization - GO:0050708 — regulation of protein secretion - GO:0008285 — negative regulation of cell population proliferation - GO:1901874 — negative regulation of post-translational protein modification - GO:0010976 — positive regulation of neuron projection development

Cellular Component: - GO:0005783 — endoplasmic reticulum - GO:0005788 — endoplasmic reticulum lumen - GO:0032991 — protein-containing complex - GO:0070062 — extracellular exosome

Molecular Profiling Data Availability

Transcriptomics/Proteomics/Metabolomics: No OI Type VIII-specific omics datasets are publicly available in GEO, PRIDE, or MetaboLights as of 2026. The extreme rarity of the disease (<1/1,000,000 globally) severely limits multi-omics profiling. General OI transcriptomics (primarily dominant forms) have identified upregulated TGF-beta signaling

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and UPR activation as key molecular signatures. Collagen biochemistry (mass spectrometry of 3-hydroxyproline levels, gel electrophoresis for overmodification) remains the primary molecular diagnostic tool specific to OI Type VIII.

Single-cell / Spatial transcriptomics: Not yet applied to OI Type VIII tissue. Single-cell studies of osteoblast differentiation in general OI models would be informative but have not been published.

7. Anatomical Structures Affected

Organ Level

Primary organs: - **Skeletal system** (UBERON:0001434): All bones affected; long bones, spine, and ribs most severely - **Bone tissue** (UBERON:0002481): Both cortical and trabecular bone affected

Secondary organ involvement: - **Eyes** (UBERON:0000970): White sclerae; rare retinal tears

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- **Ears** (UBERON:0001690): Hearing impairment (demonstrated in mouse model;
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- **Respiratory system** (UBERON:0001004): Compromised by chest deformity and scoliosis -
Vascular system (UBERON:0004537): Rare vascular malformations

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- **Teeth** (UBERON:0001091): Possible dentinogenesis imperfecta

Body systems involved: - Musculoskeletal system (primary) - Connective tissue system (primary) - Respiratory system (secondary) - Special senses (secondary)

Tissue and Cell Level

Affected tissues: - Bone tissue (UBERON:0002481) - Cartilage (UBERON:0002418) — growth plate cartilage - Connective tissue (UBERON:0002384) - Scleral tissue - Dentin

Affected cell populations: - **Osteoblasts** (CL:0000062): Primary collagen-producing cells in bone; harbor the metabolic defect - **Osteocytes** (CL:0000137): Embedded in abnormal matrix - **Chondrocytes** (CL:0000138): Growth plate dysfunction - **Fibroblasts** (CL:0000057): Skin and connective tissue involvement; demonstrate collagen overmodification in culture

Subcellular Level

- **Endoplasmic reticulum** (GO:0005783): Primary site of collagen modification and folding; enlarged ER cisternae due to collagen retention

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- **Golgi apparatus** (GO:0005794): Collagen transit affected
- **Extracellular matrix** (GO:0031012): Abnormal collagen fibrils and matrix organization

Localization

- **UBERON:0002495** (long bone): Femur, tibia, humerus most commonly fractured
- **UBERON:0001130** (vertebral column): Compression fractures, severe scoliosis
- **UBERON:0002228** (rib): Gracile ribs without beading
- **UBERON:0003129** (skull): Wormian bones, variable calvarian mineralization
- **Lateralization:** Bilateral; symmetric involvement of skeletal system

8. Temporal Development

Onset

- **Typical age of onset:** Congenital/prenatal (HP:0003577). Many cases present with intrauterine fractures detectable on prenatal ultrasound

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- **Onset pattern:** Congenital with severe manifestations from birth

- Prenatal ultrasound may show short limbs (97%), bowing of long bones (89%), cranial hypomineralization, and fractures (76%)

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Progression

Disease stages: - **Prenatal:** Intrauterine fractures, short limbs, undermineralization on ultrasound - **Neonatal:** Multiple fractures, respiratory compromise in severe cases (may be lethal) - **Infancy/early childhood:** Recurrent fractures, progressive bone deformity, failure to thrive - **Childhood/adolescence:** Severe growth deficiency, scoliosis, wheelchair dependence in most - **Adulthood:** Few patients survive to adulthood; those who do have severe osteochondrodysplasia

Progression rate: Progressive. The oldest reported patient with P3H1 deficiency was 17 7/12 years at time of report

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Disease course: Chronic, progressive, lifelong in survivors

Disease duration: Lifelong for survivors; lethal in severe cases (perinatal/neonatal death)

Patterns

- **Remission:** No spontaneous remission. Fracture frequency may decrease with bisphosphonate therapy and with increasing age/skeletal maturity.
- **Critical periods:**
 - Prenatal/perinatal: Highest risk of lethal outcome
 - Early childhood: Critical window for initiating bisphosphonate therapy
 - Growth periods: Increased fracture risk during rapid growth

9. Inheritance and Population

Epidemiology

Prevalence: - OI Type VIII is very rare globally - Recessive OI (all forms) accounts for 5-10% of all OI cases

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- P3H1 mutations are among the most common causes of AR OI in some populations

(P 40650436)

"P3H1 (n = 11) was the most frequently implicated AR gene causing OI") - Orphanet classifies OI Type VIII as ultra-rare (<1/1,000,000)

Incidence: - In West Africa (Nigeria/Ghana): Predicted 1/18,260 births based on carrier frequency

(P 22281939)

- In African Americans: Predicted 1/260,000 births

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- In other populations: Extremely rare; individual case reports

Genetic Inheritance

- **Inheritance pattern:** Autosomal recessive (HP:0000007)
- **Penetrance:** Complete in homozygotes/compound heterozygotes for null alleles
- **Expressivity:** Variable severity spectrum from lethal to severe non-lethal
- **Genetic anticipation:** Not applicable (not a repeat expansion disorder)
- **Germline mosaicism:** Not reported as a significant factor (unlike dominant OI)
- **Founder effects:**
- **West African founder (c.1080+1G>T):** Originating 650-900 years BP, shared haplotype of 63-770 Kb

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- **Bashkir population:** Ethnospecific variants p.Glu351Ter, p.Gly650Arg, c.1720+4G>A identified

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- **Consanguinity role:** Major. High rates of consanguinity significantly increase risk of homozygosity. In India, consanguinity was present in 100% of AR OI families

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- **Carrier frequency:**

- Nigeria/Ghana: 1.48%

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- African Americans: 0.4%

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- Other populations: Very low

Population Demographics

- **Affected populations:**

- West African descent (highest carrier frequency for founder mutation)
- African Americans
- Middle Eastern/North African (due to consanguinity)
- South Asian/Indian (high consanguinity rates;

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- Bashkir population of Russia

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- Reported in Brazilian, European, Egyptian cohorts

- **Geographic distribution:** Global, but concentrated in populations with high consanguinity or specific founder mutations

- **Sex ratio:** 1:1 (autosomal recessive, no sex predilection)

- **Age distribution:** Congenital; lethal cases die perinatally; survivors present in infancy/childhood

10. Diagnostics

Clinical Tests

Laboratory tests: - **Collagen biochemistry (gel electrophoresis):** Shows overmodification pattern of type I collagen

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- **Mass spectrometry:** Demonstrates nearly absent 3-hydroxylation of $\alpha 1(I)Pro986$ (1-4% vs. normal ~100%)

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- **Serum bone turnover markers:** Normal formation markers (osteocalcin, alkaline phosphatase), may have elevated resorption markers - **Calcium, phosphate, vitamin D:** Usually normal - LOINC terms: Collagen cross-links, alkaline phosphatase

Imaging: - **Radiographs:** Generalized osteopenia, undertubulation of long bones, gracile ribs without beading, Wormian bones, fractures, bowing deformities - **DXA (bone densitometry):** Extremely low areal BMD (Z-score -5 to -6 at L1-L4)

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- **Prenatal ultrasound:** Short limbs, bowing, fractures, undermineralized calvarium

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- **Micro-CT:** Research tool for detailed bone microarchitecture (research only)

Biopsy findings: - **Bone histomorphometry:** Decreased cortical width, very thin trabeculae, patches of increased osteoid

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- **QBEI (quantitative backscattered electron imaging):** Increased matrix mineralization with increased proportion of low-mineralization bone

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- **Electron microscopy (collagen fibrils):** Abnormal diameters, irregular borders

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Genetic Testing

Recommended approach: (Per EMQN best practice guidelines;

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1. Clinical suspicion of OI 2. Biochemical testing: Collagen electrophoresis showing overmodification 3. If no COL1A1/COL1A2 mutation → screen P3H1, CRTAP, PPIB 4. Whole exome sequencing (WES) increasingly used as first-line

P 34637196

Specific modalities: - **WES:** High diagnostic yield (92.3% in one cohort;

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Preferred for comprehensive screening. - **Gene panels:** OI-specific panels including P3H1, COL1A1, COL1A2, CRTAP, PPIB, SERPINF1, FKBP10, and other OI genes

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- **Single gene testing:** P3H1 sequencing when specific clinical/biochemical features suggest Type VIII - **Targeted mutation analysis:** For c.1080+1G>T in individuals of West African descent - **NGS panels:** Semiconductor sequencing panels covering COL1A1, COL1A2, CRTAP, LEPRE1/P3H1 shown to be cost-effective

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Clinical Criteria

Diagnostic criteria for OI Type VIII: 1. Severe to lethal OI phenotype with congenital fractures 2. Autosomal recessive inheritance pattern (consanguineous parents, affected siblings) 3. White sclerae (not blue) 4. Undertubulation of long bones, gracile ribs without beading 5. Collagen overmodification on biochemical testing 6. Absent/minimal Pro986 3-hydroxylation 7. Biallelic pathogenic variants in P3H1

Differential diagnosis: - **OI Type II (lethal dominant):** Blue sclerae, beaded ribs, dominant inheritance - **OI Type III (severe dominant):** Blue/grey sclerae, more variable presentation - **OI Type VII (CRTAP deficiency):** Very similar phenotype; distinguished by molecular testing - **OI Type IX (PPIB deficiency):** Similar; molecular testing differentiates - **Thanatophoric dysplasia:** Can mimic severe OI on prenatal ultrasound - **Hypophosphatasia:** Undermineralized skeleton but different biochemistry

Screening

- **Prenatal ultrasound:** Can detect severe OI (short limbs, fractures) from second trimester

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- **Carrier screening:** Population-specific screening for c.1080+1G>T in West African descent populations could be considered
- **Cascade genetic screening:** Recommended for families with known P3H1 mutations
- **Preimplantation genetic diagnosis (PGD):** Available for families with known mutations

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- **Prenatal genetic diagnosis:** Available via CVS or amniocentesis

11. Outcome/Prognosis

Survival and Mortality

- **Survival:** Ranges from lethal (perinatal death) to survival into young adulthood
- **Lethal forms:** Many patients die in the perinatal period or early infancy from respiratory insufficiency due to small thorax and rib fractures
- **Non-lethal forms:** Survival into adolescence and young adulthood is possible with aggressive management
- **Oldest reported patient:** 17 7/12 years at time of report

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- **Life expectancy:** Significantly reduced in most cases; long-term survival data are limited

Morbidity and Function

- **Mobility:** Most survivors are non-ambulatory; wheelchair dependence is common
- **Growth:** Extreme growth deficiency is universal in survivors
- **Respiratory:** Thoracic deformity and scoliosis compromise respiratory function
- **Pain:** Chronic bone pain is a significant burden
- **Quality of life:** Severely impacted across all domains (physical, social, emotional)

- **Disability:** Severe; most patients require full-time assistance

Complications

- Recurrent fractures with progressive deformity
- Severe scoliosis requiring surgical intervention
- Respiratory compromise from thoracic deformity
- Hearing loss (predicted from animal models;

P 23186870

- Retinal detachment (rare;

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- Basilar invagination (potential in severe forms)
- Chronic pain
- Immobility-related complications

Prognostic Factors

- **Mutation type:** Null mutations more severe than hypomorphic
- **Age at diagnosis:** Earlier diagnosis allows earlier treatment
- **Access to multidisciplinary care:** Significantly impacts outcomes
- **Specific phenotype at birth:** Presence of respiratory distress and rib fractures predict worse outcomes
- **Phenotypic severity ranking:** CRTAP > P3H1 > other AR genes for severity

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12. Treatment

Pharmacotherapy

Bisphosphonates (first-line medical therapy; CHEBI:77633): - **Pamidronate** (CHEBI:7903; IV, cyclic): Standard of care for moderate-to-severe OI

([P 18404382](#))

- MAXO:0001177 (bisphosphonate therapy) - Dose: 6-9 mg/kg/year in divided cycles - Effect: Increases BMD, reduces fracture rate, alleviates pain

([P 30249227](#))

- "Cyclic intravenous pamidronate is now the standard of care for moderately to severely affected children with OI"

([P 18404382](#))

- **Zoledronic acid** (CHEBI:46557; IV): Increasingly replacing pamidronate in some centers

([P 37339526](#))

- More potent; less frequent dosing (every 6 months) - Dose: 0.025-0.05 mg/kg every 6 months - "Densitometry parameters before and after zoledronic treatment were evaluated and showed significant improvement both in lumbar spine-bone mineral density Z-score and femoral neck-bone"

([P 37339526](#))

- **Oral bisphosphonates** (alendronate, risedronate): Less commonly used in severe forms

Denosumab: - RANKL antibody; used for OI forms with increased bone resorption

([P 25257953](#))

- Experience in OI Type VIII specifically is limited - May be considered for treatment-refractory cases

Calcium (CHEBI:22984) and Vitamin D (CHEBI:27300) supplementation: Supportive

Surgical and Interventional

Intramedullary rodding (primary surgical approach): - MAXO:0000004 (surgical procedure)

- **Telescoping rods** (Fassier-Duval): Preferred over static rods; longer survival times, fewer surgeries

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- **SLIM nail:** Novel device for patients with narrow medullary canals

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- Static rods: Used when telescoping rods are not feasible - Indications: Recurrent fractures, progressive deformity, facilitation of mobilization - Both patients reported by Bala & Bala

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required "multiple operations to correct both fractures and severe scoliosis"

Spinal surgery: - Posterior spinal fusion for severe scoliosis - Complex due to osteopenic bone quality

Retinal surgery: - Retinal detachment repair for giant retinal tears, with challenges due to scleral thinning

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Supportive and Rehabilitative

- **Physiotherapy:** Essential for maintaining mobility and muscle strength
- MAXO:0000011 (physical therapy)
- **Occupational therapy:** For adaptive daily living skills
- **Wheelchair/mobility aids:** Most patients require powered mobility
- **Pain management:** Multimodal approach; bisphosphonates help reduce bone pain
- **Nutritional support:** Adequate calcium, vitamin D, protein
- **Respiratory support:** May be needed for thoracic insufficiency
- **Hearing aids:** If hearing loss is present

Experimental Treatments

Anti-sclerostin antibodies (setrusumab/BPS804): - Phase 2a trial (NCT05312697) in adults with moderate OI (types I, III, IV) showed: P1NP increased 84%, BSAP increased 59%, aBMD increased 4% (p=0.038)

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- Phase 3 trial (NCT05768854) comparing setrusumab vs bisphosphonates in pediatric OI is

active (not yet recruiting type VIII specifically) - "BPS804 treatment downregulated CTX-1 by 44% from baseline...and increased aBMD by 4% (p = 0.038)"

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Anti-TGF-beta antibody (fresolimumab): - Phase I trial (NCT03064074) in 8 adults with OI

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- TGF-beta pathway identified as top activated signaling pathway in OI bone; SMAD phosphorylation was the most significantly upregulated GO molecular event - **Critical finding for OI Type VIII:** "Treatment with fresolimumab was well-tolerated and associated with increases in LS aBMD in participants with OI type IV, whereas participants with OI type III and VIII had unchanged or decreased LS aBMD"

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- This differential response suggests recessive OI may require different therapeutic approaches than dominant forms

Stem cell/mesenchymal stromal cell transplantation: - Allogeneic bone marrow transplant in 3 children with OI showed 1.5-2.0% donor osteoblast engraftment, new dense bone formation, and increased growth velocity

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- BOOSTB4 prenatal stem cell therapy: Phase 1/2 trial (NCT03706482) — mesenchymal stem cells delivered prenatally; active, not yet recruiting - Postnatal BOOST cells: Phase 1/2 trial (NCT04623606) — status unknown - Not yet tested specifically in OI Type VIII

Combination therapy (PTH + bisphosphonate): - NCT03735537: Teriparatide (PTH analog, anabolic) + zoledronic acid in OI — completed - NCT01679080: Teriparatide + zoledronic acid — terminated

4-PBA (4-phenylbutyrate; CHEBI:8104): - Chemical chaperone shown to improve osteoblast homeostasis in OI mouse models

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- Preclinical stage; not yet in clinical trials for OI

Gene therapy: - Research-stage for OI generally; no specific trials for OI Type VIII - CRISPR-based correction of collagen mutations explored in cell models - A 2025 review of gene editing for collagen disorders

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highlights CRISPR-Cas9, base editing, and prime editing as "promising therapeutic options for these disorders, representing a putative one-for-all treatment strategy," though challenges include "the lack of recurring mutations" in OI — a particular barrier for recessive OI Type VIII where most families carry private mutations

Treatment Strategy

Multidisciplinary approach required: 1. Pediatric endocrinology/metabolic bone specialist (bisphosphonate management) 2. Orthopedic surgery (fracture management, rodding, scoliosis correction) 3. Physiotherapy and rehabilitation 4. Pain management 5. Ophthalmology (retinal surveillance) 6. Audiology (hearing monitoring) 7. Respiratory medicine (for thoracic insufficiency) 8. Genetic counseling

Treatment outcomes: - Bisphosphonate therapy: Significant reduction in fracture rate (from 1.5 to 0.7 fractures/year;

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- Bisphosphonates increase LS BMD significantly - Combined with intramedullary rodding, enables improved function - Long-term benefits maintained with maintenance dosing

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13. Prevention

Primary Prevention

- **Genetic counseling:** Essential for families with known P3H1 mutations or carrier status
- MAXO:0000079 (genetic counseling)
- **Carrier screening:** Targeted screening for c.1080+1G>T in individuals of West African/African American descent
- **Preconception counseling:** For consanguineous couples from high-risk populations
- **Avoidance of consanguinity:** Reduces risk in populations with elevated carrier frequency

Secondary Prevention (Early Detection)

- **Prenatal ultrasonography:** Can detect severe OI features (short limbs, fractures, undermineralization) from ~16-20 weeks gestation
- **Prenatal molecular diagnosis:** Available via CVS or amniocentesis when familial mutation is known

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- **Preimplantation genetic diagnosis (PGD):** For affected families using IVF
- **Newborn/neonatal screening:** Not part of routine newborn screening programs; clinical suspicion based on fractures and radiographic findings

Tertiary Prevention

- **Bisphosphonate therapy:** Reduces fracture frequency and bone pain
- **Intramedullary rodding:** Prevents recurrent fractures and progressive deformity
- **Fall prevention:** Environmental modifications and mobility aids
- **Careful handling:** Especially in infancy (fracture prevention protocols)
- **Respiratory monitoring:** Early intervention for thoracic insufficiency
- **Ophthalmologic surveillance:** For early detection of retinal pathology

Public Health

- Population carrier screening for the West African founder mutation could be considered in endemic regions, given the 1.48% carrier frequency and predicted 1/18,260 affected births

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- Education of healthcare providers about recessive OI in high-carrier-frequency populations

14. Other Species / Natural Disease

Naturally Occurring Disease

No naturally occurring OI Type VIII (P3H1 deficiency) has been specifically described in other species. However, osteogenesis imperfecta-like conditions occur naturally in several species:

- **Dogs** (NCBI Taxon: 9615, *Canis lupus familiaris*): Naturally occurring OI described in multiple breeds
- OMIA: 000720 (Osteogenesis imperfecta in *Canis lupus familiaris*)
- **Cattle** (NCBI Taxon: 9913, *Bos taurus*): OI described in several breeds
- OMIA: 000720 (Osteogenesis imperfecta in *Bos taurus*)
- **Cats** (NCBI Taxon: 9685, *Felis catus*): Rare cases reported

Comparative Biology

- P3H1 is an evolutionarily conserved gene present in vertebrates
- Orthologous gene in mouse: *Lepre1* (NCBI Gene: 56401)
- Orthologous gene in zebrafish: *p3h1* (NCBI Gene: 569876)
- Notably, wild-type zebrafish lack collagen type I 3-hydroxyproline, providing a unique model to study the chaperone vs. enzymatic function of the complex

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Zoonotic Potential

Not applicable. OI Type VIII is a genetic disorder with no infectious or zoonotic component.

15. Model Organisms

Mouse Models

P3H1 null mouse (*Lepre1* knockout): - **Type:** Genetic knockout (loss of function) - **NCBI Taxon:** 10090 (*Mus musculus*) - **Gene ID:** 56401 (NCBI Gene); MGI:1926146 (*Lepre1/P3h1*) - **Phenotype recapitulation:** - OI-like skeletal phenotype with bone fragility - Hearing

impairment: 20-30 dB increased ABR thresholds

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- Abnormal morphology of middle ear bone joints (incudostapedial and incudomalleal) - Growth deficiency - **Applications:** Study of hearing loss mechanism; bone biology; therapeutic testing - **Limitations:** Mouse bone remodeling differs from human; severity may not fully recapitulate human lethal forms

P3H1 catalytic-dead knock-in mouse (*Lepre1*^{H662A}): - **Type:** Knock-in (single amino acid substitution abolishing enzymatic activity while retaining complex formation) - Abolishes 3-hydroxylation at Pro986 but retains CRTAP binding and chaperone function

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- **Key finding:** Milder phenotype than null, indicating chaperone function is critical for severity - **Applications:** Dissecting enzymatic vs. chaperone functions of the complex

CyPB null mouse (*Ppib* knockout): - Develops kyphosis and severe osteoporosis

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- Abnormal collagen fibril morphology - P3H1 levels substantially reduced; CRTAP unaffected - **Applications:** Study of CyPB role in the prolyl 3-hydroxylation complex

Zebrafish Models

p3h1 and crtap CRISPR/Cas9 knockouts: - **NCBI Taxon:** 7955 (*Danio rerio*) - **Gene IDs:** 569876 (*p3h1*); 558836 (*crtap*) - **Phenotype** **recapitulation**

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- Reduced size and body disproportion - Altered mineralization - Vertebral body fusions, deformities, and fractures - Reduced bone size, thickness, and bone volume - Intracellular collagen overmodification and ER retention (enlarged ER cisternae) - Extracellular disorganized collagen fibers with altered diameter - **Unique advantage:** Wild-type zebrafish lack collagen 3-hydroxyproline, allowing isolation of chaperone function from enzymatic function - **Key conclusion:** "The data support the defective chaperone role of the 3-hydroxylation complex as the primary cause of the skeletal phenotype"

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Model Resources

- MGI (Mouse Genome Informatics): *Lepre1* alleles
 - ZFIN (Zebrafish Information Network): *p3h1* alleles
 - IMPC (International Mouse Phenotyping Consortium): Phenotype data
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Summary

Osteogenesis Imperfecta Type VIII is a severe-to-lethal autosomal recessive bone fragility disorder caused by biallelic loss-of-function mutations in the *P3H1* gene (formerly *LEPRE1*). The disease results from loss of the prolyl 3-hydroxylation complex (P3H1/CRTAP/CyPB) in the endoplasmic reticulum, leading to defective collagen post-translational modification and impaired chaperone-assisted folding. Key distinguishing features from dominant OI include white sclerae, undertubulation of long bones, gracile ribs without beading, and extreme growth deficiency. A West African founder mutation (c.1080+1G>T) with 1.48% carrier frequency in Nigeria/Ghana makes this disease particularly relevant in West African and African American populations. Treatment is currently limited to bisphosphonate therapy combined with orthopedic management (intramedullary rodding), physiotherapy, and multidisciplinary supportive care. Mouse and zebrafish models have confirmed that the chaperone function of the P3H1/CRTAP complex — not the enzymatic 3-hydroxylation activity alone — is the primary driver of the skeletal phenotype.

Consolidated Ontology Reference

Disease and Phenotype Ontologies

Ontology	ID	Term
MONDO	MONDO:0012581	Osteogenesis imperfecta type VIII
OMIM	610915	Osteogenesis imperfecta, type VIII
OMIM	610339	P3H1 gene
HP	HP:0005855	Multiple prenatal fractures
HP	HP:0002757	Recurrent fractures
HP	HP:0000938	Osteopenia
HP	HP:0008873	Disproportionate short-limb short stature
HP	HP:0000883	Thin ribs
HP	HP:0002953	Vertebral compression fracture
HP	HP:0000260	Wide anterior fontanel
HP	HP:0001552	Barrel-shaped chest
HP	HP:0000311	Round face
HP	HP:0000592	White sclerae (blue sclerae absent)
HP	HP:0000007	Autosomal recessive inheritance
HP	HP:0003577	Congenital onset

Gene Ontology (GO) Terms

GO ID	Term	Category
GO:0019797	Procollagen-proline 3-dioxygenase activity	Molecular function
GO:0005506	Iron ion binding	Molecular function
GO:0031418	L-ascorbic acid binding	Molecular function
GO:0060348	Bone development	Biological process
GO:0032963	Collagen metabolic process	Biological process
GO:0006457	Protein folding	Biological process
GO:0018126	Protein hydroxylation	Biological process
GO:0030199	Collagen fibril organization	Biological process
GO:0030968	Unfolded protein response	Biological process
GO:0005783	Endoplasmic reticulum	Cellular component
GO:0031012	Extracellular matrix	Cellular component

Anatomical Ontology (UBERON) and Cell Ontology (CL)

Ontology	ID	Term
UBERON	UBERON:0001434	Skeletal system
UBERON	UBERON:0002481	Bone tissue
UBERON	UBERON:0002495	Long bone
UBERON	UBERON:0001130	Vertebral column
UBERON	UBERON:0002228	Rib
UBERON	UBERON:0000970	Eye
CL	CL:0000062	Osteoblast
CL	CL:0000137	Osteocyte
CL	CL:0000138	Chondrocyte
CL	CL:0000057	Fibroblast

Chemical Entities (CHEBI)

CHEBI ID	Term	Role in OI Type VIII
CHEBI:29033	Fe ²⁺ (iron(2+))	P3H1 catalytic cofactor
CHEBI:29073	L-ascorbate (vitamin C)	P3H1 catalytic cofactor
CHEBI:16810	2-oxoglutarate	P3H1 co-substrate
CHEBI:15741	Succinate	P3H1 reaction product
CHEBI:18240	3-hydroxy-L-proline	Product of P3H1 enzymatic activity; absent in OI VIII
CHEBI:4031	Cyclosporin A	CyPB/PPIB inhibitor; used in structural studies
CHEBI:77633	Bisphosphonate	First-line drug class for treatment
CHEBI:7903	Pamidronate	Standard IV bisphosphonate therapy
CHEBI:46557	Zoledronic acid	Alternative IV bisphosphonate therapy
CHEBI:8104	4-phenylbutyrate	Experimental chemical chaperone
CHEBI:22984	Calcium	Nutritional supplement
CHEBI:27300	Vitamin D	Nutritional supplement

Medical Action Ontology (MAXO)

MAXO ID	Term	Application
MAXO:0001177	Bisphosphonate therapy	First-line pharmacotherapy
MAXO:0000004	Surgical procedure	Intramedullary rodding, spinal fusion
MAXO:0000011	Physical therapy	Rehabilitation and mobility
MAXO:0000079	Genetic counseling	Prevention and family planning

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